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END TERM EXAMINATION
FIRST SEMESTER [B.TECH] DECEMBER 2025

Paper Code: BS-111

Subject: Applied Mathematics-I

Time: 3 Hours

Maximum Marks: 60

Note: Attempt all questions as directed. Internal choice is indicated.

Q. No.	Question	Marks
Q.1	Attempt any four of the following questions:	(4 × 5 = 20)
(a)	If $z = x^y + y^x$, then prove that $(\partial^2 z / \partial x \partial y) = (\partial^2 z / \partial y \partial x)$.	5
(b)	Find the value of: $(d/dx) \int_{x^2}^{x^3} (1 / \log t) dt$.	5
(c)	Solve the differential equation: $(1 + y^2) dx = (\tan^{-1} y - x) dy$.	5
(d)	Express $4x^3 + 6x^2 + 7x + 2$ in terms of Legendre's polynomials.	5
(e)	Convert given Matrix A into normal form and find its rank: $A = [[1, 2, 1], [-1, 0, 2], [2, 1, -3]]$ [Rows: [1, 2, 1], [-1, 0, 2], [2, 1, -3]]	5
(f)	Examine the following vectors for linear dependence and find the relation if it exists: $X_1 = (1, 2, 4), X_2 = (2, -1, 3), X_3 = (0, 1, 2), X_4 = (-3, 7, 2)$.	5
(g)	Determine the constants a, b, c so that: $F = (x + 2y + az) \hat{i} + (bx - 3y - z) \hat{j} + (4x + cy + 2z) \hat{k}$ is irrotational.	5
(h)	Show that $\text{div}(\text{grad } r^n) = n(n + 1)r^{n-2}$.	5
Q.2	If $W = f(x, y), x = r \cos \theta, y = r \sin \theta$, show that:	5
(a)	$(\partial f / \partial x)^2 + (\partial f / \partial y)^2 = (\partial W / \partial r)^2 + (1 / r^2) (\partial W / \partial \theta)^2$.	
(b)	If $x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$, find the Jacobian $\partial(x, y, z) / \partial(r, \theta, \phi)$.	5
	OR	
Q.3	In a plane triangle ABC , find the maximum value of $\cos A \cos B \cos C$.	5
(a)		
(b)	If $V = f(2x - 3y, 3y - 4z, 4z - 2x)$, compute the value of $6V_x + 4V_y + 3V_z$.	5

Q. No.	Question	Marks
Q.4 (a)	Solve the differential equation: $(d^2y/dx^2) - 2 (dy/dx) + y = e^x x \cos x$.	5
(b)	Find the orthogonal trajectories of the family of parabolas $y^2 = 4a(x + a)$.	5
	OR	
Q.5 (a)	Solve $(d^2y/dx^2) - y = (1 + e^{-x})^{-2}$ by the method of variation of parameters.	5
(b)	Prove that $J_0(0) = 1$ for Bessel function of the first kind.	5
Q.6 (a)	Investigate the values of a and b so that the system of equations: $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, $2x + 3y + az = b$ have: (i) No solution, (ii) A unique solution, (iii) An infinite number of solutions.	5
(b)	Solve the system of linear equations by Gauss elimination method: $x + 3y + z = 10$, $x - 2y - z = -6$, $2x + y + z = 10$.	5
	OR	
Q.7 (a)	Find the eigenvalues and eigenvectors of the matrix: $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ [Rows: [8, -6, 2], [-6, 7, -4], [2, -4, 3]]	5
(b)	Verify Cayley-Hamilton theorem for the matrix: $A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$. Hence find A^{-1} .	5
Q.8	Verify Stoke's theorem for the function $F = x^2 \hat{i} - xy \hat{j}$, integrated round the square in the plane $z = 0$ and bounded by the lines $x = 0$, $y = 0$, $x = a$, $y = a$.	10
	OR	
Q.9 (a)	Find the arc length of the semi-cubical parabola $\vec{r}(t) = (t \hat{i} + t^{3/2} \hat{j})$ from $(0, 0, 0)$ to $(4, 8, 0)$.	5
(b)	Find the directional derivative of $1/r$ in the direction $\vec{r}$, where $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$.	5

**** End of Question Paper ****